실습과제 2

```
linux@BOOK-M8LSJ7JPPD: $ source /opt/ros/jazzy/setup.bash
linux@BOOK-M8LSJ7JPPD: $ source ~/ros2_ws/install/local_setup.bash
linux@BOOK-M8LSJ7JPPD: $ cd ~/ros2_ws/src
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ ros2 pkg create pub2_2 --build-type ament_cmake --dependencies rclcpp geometry_msgs

Aborted!
The directory already exists: ./pub2_2
Either remove the directory or choose a different destination directory or package name
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ tree pub2_2
pub2_2
├── CMakeLists.txt
├── include
│   └── pub2_2
├── package.xml
└── src

4 directories, 2 files
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$
```

```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ nano pub2_2/src/pub.cpp
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$
```

```
 GNU nano 7.2                                  pub2_2/src/pub.cpp
#include "rclcpp/rclcpp.hpp"
#include "geometry_msgs/msg/vector3.hpp"
#include <memory>
#include <chrono>
#include <functional>
#include <iostream>

using namespace std::chrono_literals;

// 타이머 콜백 함수
void callback(
    rclcpp::Node::SharedPtr node,
    rclcpp::Publisher<geometry_msgs::msg::Vector3>::SharedPtr mypub,
    double x, double y, double z)
{
    auto message = geometry_msgs::msg::Vector3();

    message.x = x;
    message.y = y;
    message.z = z;

    RCLCPP_INFO(node->get_logger(),
                "Publish: x=%.2f y=%.2f z=%.2f",
                message.x, message.y, message.z);

    mypub->publish(message);
}

int main(int argc, char* argv[])
{
    rclcpp::init(argc, argv);

    auto node = std::make_shared<rclcpp::Node>("mynode");

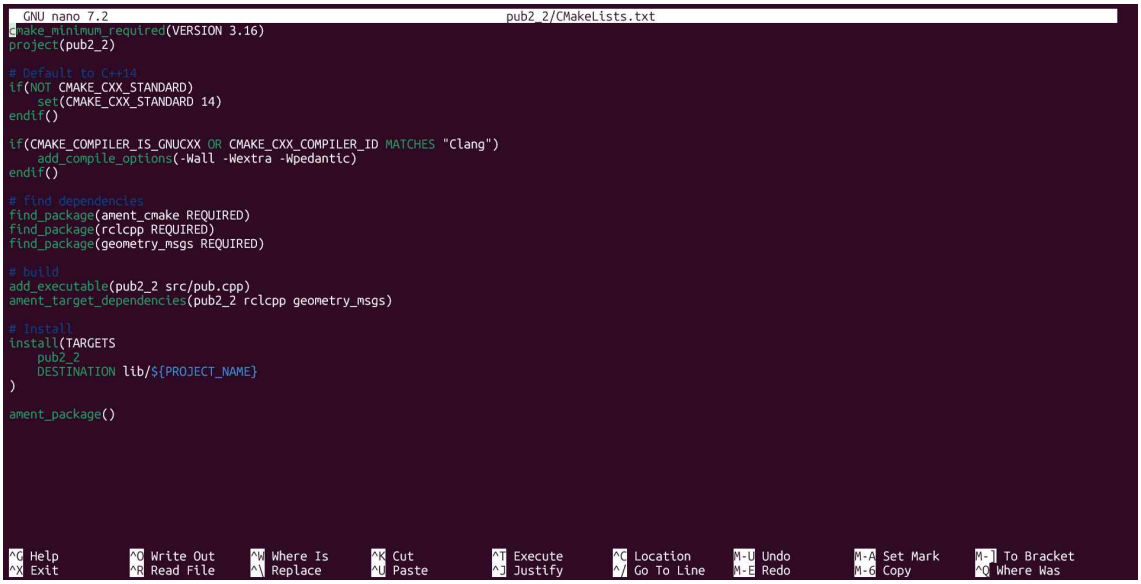
    auto qos_profile = rclcpp::QoS(rclcpp::KeepLast(10));

^G Help          ^O Write Out     ^W Where Is      ^K Cut           ^T Execute       ^C Location      M-U Undo         M-A Set Mark     M-] To Bracket
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line    M-E Redo         M-6 Copy         ^Q Where Was
```

```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ tree pub2_2
pub2_2
├── CMakeLists.txt
├── include
│   └── pub2_2
├── package.xml
└── src
    └── pub.cpp

4 directories, 3 files
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$
```

```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ nano pub2_2/CMakeLists.txt
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$
```


GNU nano 7.2 pub2_2/CMakeLists.txt
cmake_minimum_required(VERSION 3.16)
project(pub2_2)
Default to C++14
if(NOT CMAKE_CXX_STANDARD)
set(CMAKE_CXX_STANDARD 14)
endif()
if(CMAKE_COMPILER_IS_GNUCXX OR CMAKE_CXX_COMPILER_ID MATCHES "Clang")
add_compile_options(-Wall -Wextra -Wpedantic)
endif()
find dependencies
find_package(ament_cmake REQUIRED)
find_package(rclcpp REQUIRED)
find_package(geometry_msgs REQUIRED)
build
add_executable(pub2_2 src/pub.cpp)
ament_target_dependencies(pub2_2 rclcpp geometry_msgs)
Install
install(TARGETS
pub2_2
DESTINATION lib/${PROJECT_NAME}
)
ament_package()
^C Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ cat pub2_2/CMakeLists.txt
cmake_minimum_required(VERSION 3.16)
project(pub2_2)

# Default to C++14
if(NOT CMAKE_CXX_STANDARD)
    set(CMAKE_CXX_STANDARD 14)
endif()

if(CMAKE_COMPILER_IS_GNUCXX OR CMAKE_CXX_COMPILER_ID MATCHES "Clang")
    add_compile_options(-Wall -Wextra -Wpedantic)
endif()

# find dependencies
find_package(ament_cmake REQUIRED)
find_package(rclcpp REQUIRED)
find_package(geometry_msgs REQUIRED)

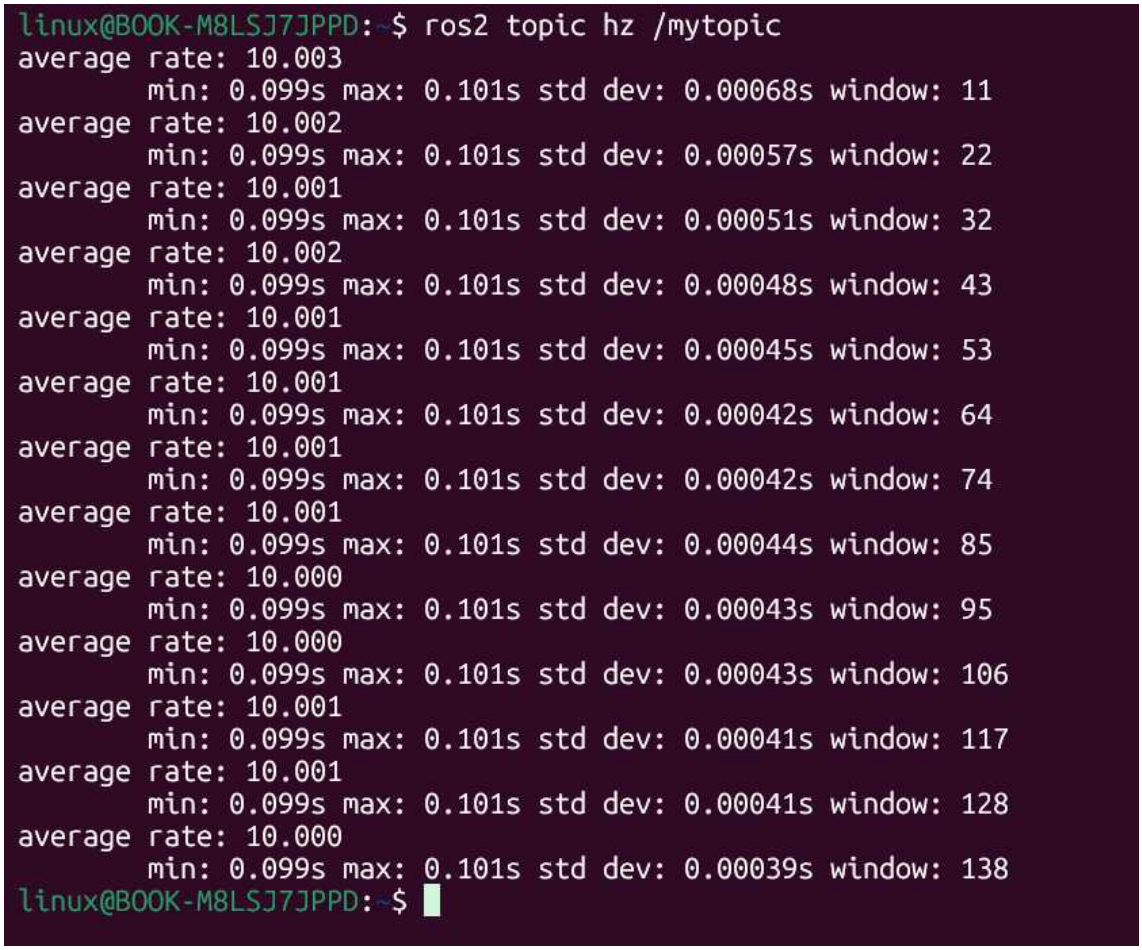
# build
add_executable(pub2_2 src/pub.cpp)
ament_target_dependencies(pub2_2 rclcpp geometry_msgs)

# Install
install(TARGETS
    pub2_2
    DESTINATION lib/${PROJECT_NAME}
)

ament_package()
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ cd ~/ros2_ws
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ colcon build --symlink-install --packages-select pub2_2
Starting >>> pub2_2
Finished <<< pub2_2 [12.9s]

Summary: 1 package finished [13.3s]
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ source install/local_setup.bash
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ ros2 run pub2_2 pub2_2
실수 3개 입력 (x y z): 1.5 2.5 3.5
[INFO] [1793605866.205242424] [mynode]: Publish: x=1.50 y=2.50 z=3.50
[INFO] [1793605866.305154403] [mynode]: Publish: x=1.50 y=2.50 z=3.50
[INFO] [1793605866.404971908] [mynode]: Publish: x=1.50 y=2.50 z=3.50
[INFO] [1793605866.505090366] [mynode]: Publish: x=1.50 y=2.50 z=3.50
[INFO] [1793605866.605155248] [mynode]: Publish: x=1.50 y=2.50 z=3.50
[INFO] [1793605866.705100604] [mynode]: Publish: x=1.50 y=2.50 z=3.50
[INFO] [1793605866.805202337] [mynode]: Publish: x=1.50 y=2.50 z=3.50
[INFO] [1793605866.905200983] [mynode]: Publish: x=1.50 y=2.50 z=3.50
[INFO] [1793605867.005009804] [mynode]: Publish: x=1.50 y=2.50 z=3.50
[INFO] [1793605867.105025579] [mynode]: Publish: x=1.50 y=2.50 z=3.50
[INFO] [1793605867.205059628] [mynode]: Publish: x=1.50 y=2.50 z=3.50
[INFO] [1793605867.305103048] [mynode]: Publish: x=1.50 y=2.50 z=3.50
```

```
linux@BOOK-M8LSJ7JPPD:~$ source /opt/ros/jazzy/setup.bash
linux@BOOK-M8LSJ7JPPD:~$ source ~/ros2_ws/install/local_setup.bash
linux@BOOK-M8LSJ7JPPD:~$ ros2 node list
/mynode
linux@BOOK-M8LSJ7JPPD:~$ ros2 topic list -t
/mytopic [geometry_msgs/msg/Vector3]
/parameter_events [rcl_interfaces/msg/ParameterEvent]
/rosout [rcl_interfaces/msg/Log]
linux@BOOK-M8LSJ7JPPD:~$ ros2 topic echo /mytopic
x: 1.5
y: 2.5
z: 3.5
---
x: 1.5
y: 2.5
z: 3.5
---
x: 1.5
y: 2.5
z: 3.5
---
x: 1.5
y: 2.5
z: 3.5
---
x: 1.5
y: 2.5
z: 3.5
---
x: 1.5
y: 2.5
z: 3.5
---
x: 1.5
y: 2.5
z: 3.5
---
x: 1.5
y: 2.5
z: 3.5
---
x: 1.5
y: 2.5
z: 3.5
```


linux@BOOK-M8LSJ7JPPD:~$ ros2 topic hz /mytopic
average rate: 10.003
min: 0.099s max: 0.101s std dev: 0.00068s window: 11
average rate: 10.002
min: 0.099s max: 0.101s std dev: 0.00057s window: 22
average rate: 10.001
min: 0.099s max: 0.101s std dev: 0.00051s window: 32
average rate: 10.002
min: 0.099s max: 0.101s std dev: 0.00048s window: 43
average rate: 10.001
min: 0.099s max: 0.101s std dev: 0.00045s window: 53
average rate: 10.001
min: 0.099s max: 0.101s std dev: 0.00042s window: 64
average rate: 10.001
min: 0.099s max: 0.101s std dev: 0.00042s window: 74
average rate: 10.001
min: 0.099s max: 0.101s std dev: 0.00044s window: 85
average rate: 10.000
min: 0.099s max: 0.101s std dev: 0.00043s window: 95
average rate: 10.000
min: 0.099s max: 0.101s std dev: 0.00043s window: 106
average rate: 10.001
min: 0.099s max: 0.101s std dev: 0.00041s window: 117
average rate: 10.001
min: 0.099s max: 0.101s std dev: 0.00041s window: 128
average rate: 10.000
min: 0.099s max: 0.101s std dev: 0.00039s window: 138
linux@BOOK-M8LSJ7JPPD:~$

```
linux@BOOK-M8LSJ7JPPD:~$ ros2 topic bw /mytopic
WARNING: topic [/mytopic] does not appear to be published yet
Subscribed to [/mytopic]
289 B/s from 10 messages
        Message size mean: 28 B min: 28 B max: 28 B
284 B/s from 20 messages
        Message size mean: 28 B min: 28 B max: 28 B
283 B/s from 30 messages
        Message size mean: 28 B min: 28 B max: 28 B
282 B/s from 40 messages
        Message size mean: 28 B min: 28 B max: 28 B
282 B/s from 50 messages
        Message size mean: 28 B min: 28 B max: 28 B
281 B/s from 60 messages
        Message size mean: 28 B min: 28 B max: 28 B
281 B/s from 70 messages
        Message size mean: 28 B min: 28 B max: 28 B
281 B/s from 80 messages
        Message size mean: 28 B min: 28 B max: 28 B
281 B/s from 90 messages
        Message size mean: 28 B min: 28 B max: 28 B
281 B/s from 100 messages
        Message size mean: 28 B min: 28 B max: 28 B
linux@BOOK-M8LSJ7JPPD:~$
```

```
linux@BOOK-M8LSJ7JPPD:~$ rqt_graph
```

